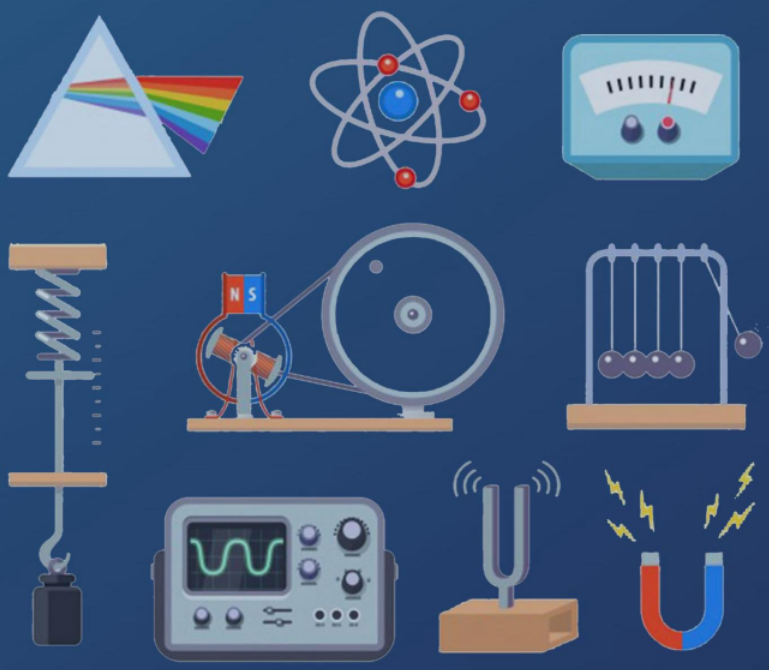
PHYSICS

Dual Nature of Radiation & Matter



SHEET

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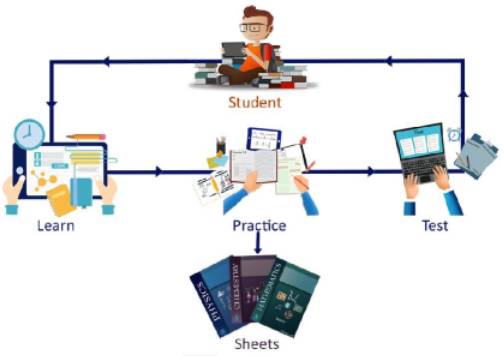
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Phtotelectric Effect and Einstein's Equation	1-8	1-5,8-12	1-9	1-7,8-13, 17-23
2	Wave nature of Matter - De Broglie Wavelength	9-17	6-13	10-15	14-16,24
3	After Full Chapter		7	16	

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Photoelectric Effect and Einstein's Equation

Q1. Who discovered photo electric effect ?

(A) Hertz (B) Lenard

(C) Hallwach (D) Einstein

Q2. The wavelength of X-ray photon is 0.01 Å, its momentum in kg m/s is -

(A) 6.6×10^{-22} (B) 6.6×10^{-32}

(C) 6.6×10^{-46} (D) 6.6×10^{-27}

Q3. The work- function of a substance is 4.0 eV. The longest wavelength of light that can cause photoelectron emission from this substance is approximately :

(A) 540 nm (B) 400 nm

(C) 310 nm (D) 220 nm

Q4. When light of wavelength lesser than 6000 Å is incident on a metal, electrons are emitted. The approximate work-function of the metal is :

(A) 1 eV (B) 2 eV

(C) 4 eV (D) 6 eV

Q5. The maximum wavelength of light for photoelectric effect from a metal is 200 nm. The maximum kinetic energy of electron which is emitted by the radiation of wave length 100 nm will be :

(A) 12.4 eV (B) 6.2 eV

(C) 100 eV (D) 200 eV

Q6. If n_r and n_b are respectively, the number of photons emitted by a red bulb and a blue bulb of equal power in a given time, then

(A) $n_r = n_b$

(B) $n_r < n_b$

(C) $n_r > n_b$

(D) nothing can be predicted

Q7. **Statement 1** : Photoelectric effect establishes quantum nature of light.

Statement 2 : There is negligible time lag between photon collisions with the material and photoelectron emission irrespective of intensity of incident light. (Assume incident light is of frequency greater than threshold frequency of the material).

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q8. Two monochromatic light sources. A and B emit the same number of photons per second. The wavelength of A is $\lambda_A = 400$ nm and that of B is $\lambda_B = 600$ nm. The power radiated by source B is -

(A) Equal to that of source A.

(B) Less than that of source A.

(C) Greater than that of source A.

(D) Cannot be compared to that from source A using the available data.

Wave Nature of Matter-De Broglie wavelength

Q9. Electron has energy of 100 eV. What will be its wavelength –

(A) 1.2 Å (B) 10 Å

(C) 100 Å (D) 1 Å

Q10. An electron is moving with velocity 6.6 x 10^3 m/s. The De-Broglie wavelength associated with electron is
(mass of electron = 9×10^{-31} Kg,
Planck's Constant = 6.62×10^{-34} J-s)
(A) 1×10^{-19} m
(B) 1×10^{-5} m
(C) 1×10^{-7} m
(D) 1×10^{-10} m

Q11. The de Broglie wavelength associated with a ball of mass 1 kg having kinetic energy 0.5 J is ?
(A) 6.626×10^{-32} cm
(B) 5.257×10^{-36} cm
(C) 6.297×10^{-36} cm
(D) 7.287×10^{-36} cm

Q12. The De-Broglie wavelength associated with electrons revolving round the nucleus in a hydrogen atom in ground state, will be :
(A) 0.3 Å (B) 3.3 Å
(C) 6.62 Å (D) 10 Å

Q13. A proton moves on a circular path of radius 6.6×10^{-3} m in a perpendicular magnetic field of 0.625 tesla. The De broglie wavlength associated with the proton will be :
(A) 1Å (B) 0.1Å
(C) 0.01Å (D) 0.001Å

Q14. de Broglie equation for an electron shows its -
(A) particle nature
(B) wave nature
(C) dual nature
(D) none of these

Q15. What is the de broglie wavelength in (Å) of the α-particle accelerated through a potential difference V.
(A) $\frac{0.287}{\sqrt{V}}$ (B) $\frac{12.27}{\sqrt{V}}$
(C) $\frac{0.101}{\sqrt{V}}$ (D) $\frac{0.22}{\sqrt{V}}$

Q16. Linear momenta of a proton and an electron are equal. Relative to an electron-
(A) Kinetic energy of proton is more.
(B) de-Broglie wavelength of proton is more.
(C) de-Broglie wavelength of proton is less.
(D) de-Broglie wavelength of proton and electron are equal.

Q17. A proton and an α - particle accelerated through same voltage. The ratio of their De-broglie wavelength will be :
(A) 1 : 2 (B) $2\sqrt{2}$:1
(C) $\sqrt{2}$:1 (D) 2 : 1

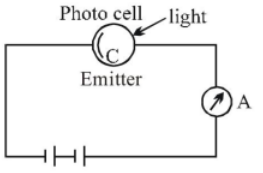
Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The stopping potential for photo electron emitted from a metal surface of work function 1.7 eV is 10.4V. Select correct choice
(A) The wavelength of light used is 1024 Å
(B) The wavelength of light used is 970.6 Å
(C) The light used may be emitted by hydrogen gas sample which de–excites from n=3 to n=1
(D) The light used may be emitted by hydrogen gas sample which de–excites from n = 2 to n = 1

Q2. The figure shows a photo cell circuit. The cathode of the photo cell is illuminated by a monochromatic light. If the intensity is kept constant and the frequency of the incident light is increased, then the



(A) Photo electric current in the circuit increases.

(B) Saturation electric current in the circuit decreases.

(C) Maximum kinetic energy of the photo electrons increases.

(D) Photo electric current in the circuit can be reduced to zero.

when the polarity of the terminals is reversed.

Q3. In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will becomes

(A) $v\sqrt{\frac{3}{4}}$

(B) $v\sqrt{\frac{4}{3}}$

(C) less than $v\sqrt{\frac{3}{4}}$

(D) greater than $v\sqrt{\frac{4}{3}}$

Q4. Photoelectric effect supports quantum nature of light because :

(A) There is a minimum frequency below which no photoelectrons are emitted.

(B) The maximum kinetic energy of photo-electrons depends only on the frequency of light and not on its intensity.

(C) Even when the metal surface is faintly illuminated the photoelectrons if any of them leave the surface immediately.

(D) Electric charge of the photoelectrons is quantized.

Q5. In photoelectric effect, stopping potential depends on -

(A) frequency of the incident light

(B) intensity of the incident light by varies source distance

(C) emitter's properties

(D) frequency and intensity of the incident light

Q6. If particles are moving with same velocity, then maximum de-Broglie wavelength will be for-

(A) proton

(B) α - particle

(C) neutron

(D) β - particle

Q7. Which of the following phenomena can be explained only on the basis of quantum theory of light?

(A) Energy spectrum of black body radiation

(B) Atomic spectra

(C) Photoelectric effect

(D) Doppler effect

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8. Some quantities related to the photoelectric effect are mentioned under Column I and Column II. Match each quantity in Column I with the corresponding quantities in Column II on which it depends

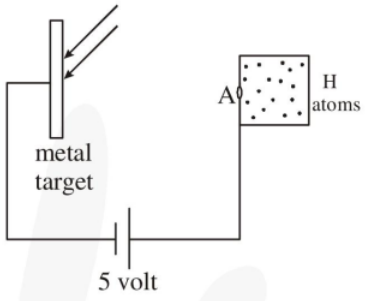
Column-I	Column-II
(A) Saturation current	(P) Frequency of light
(B) Stopping potential	(Q) Work function
(C) de-Broglie wavelength of photoelectron	(R) Area of photosensitive plate
(D) Force due to radiation falling on the photo–plate.	(S) Intensity of light (at constant frequency)
	(T) None of these

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q9. If light of wavelength 4000 Å falls on a metal which has a stopping potential 1·4 volt against photoelectric emission then what is the work function of the metal. [Take $h = 6{\cdot}6 \times 10^{-34}$ Js and $c = 3 \times 10^{8}$ ms^{-1}] (in ev).

Q10. Electromagnetic waves of wavelength 1242 Å are incident on a metal of work function 2eV. The target metal is connected to a 5 volt cell, as shown. The electrons pass through hole A into a gas of hydrogen atoms in their ground state. Find the number of spectral lines emitted when hydrogen atoms come back to their ground states after having been excited by the electrons. Assume all excitations in H-atoms from ground state only. (hc = 12420 eVÅ)



SUBJECTIVE TYPE

Q11. The work function of a metal is 4 eV. If 5000Å wavelength of light is incident on the metal, is there any photo electric effect?

Q12. A TV station is operated at 100 MW with a signal frequency of 10 MHz. Calculate the number of photons radiated per second by its antenna.

Q13. A particle of mass M at rest explodes into two particles of masses m_1 and m_2 having non zero velocities. Find out the ratio of the de–Broglie wavelengths of the two particles.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The number of photons emitted in one second by a 100 watt red light source, assuming for simplicity the average wavelength for each photon is 694 nm is-

(A) 3.49×10^{20} (B) 4.49×10^{20}

(C) 3.49×10^{18} (D) 4.49×10^{18}

Q2. Which one of the following is true in photoelectric emission-

(A) Photoelectric current is directly proportional to the amplitude of light of a given frequency.

(B) Photoelectric current is directly proportional to the intensity of light of a given frequency above threshold value.

(C) Above the threshold frequency, the maximum K.E. of photoelectrons is inversely proportional to the frequency of incident light.

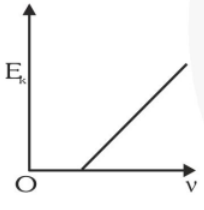
(D) The threshold frequency depends upon the wavelength of incident light.

Q3. The work functions of tungsten and sodium are 5.06 eV and 2.53 eV respectively. If the threshold wavelength for sodium is 5896 Å, then the threshold wavelength for the tungsten will be :

(A) 11792 Å (B) 5896 Å

(C) 4312 Å (D) 2948 Å

Q4. Graph is plotted between maximum kinetic energy of electron with frequency of incident photon in Photo electric effect. The slope of curve will be-



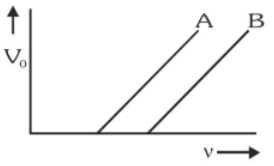
(A) charge of electron

(B) work function of metal

(C) planck's constant

(D) ratio of Planck constant and charge of electron

Q5. The stopping potential as a function of frequency of incident radiation is plotted for two different surfaces A and B. The graphs show that the work function of A is-



(A) Greater than that of B

(B) Smaller than that of B

(C) Same as that of B

(D) No comparison can be done from given graphs

Q6. The work functions for metals A, B and C are respectively 1.92 eV, 2.0 eV and 5 eV. According to Einstein's equation, the metals which will emit photo electrons for a radiation of wavelength 4100 Å is /are-

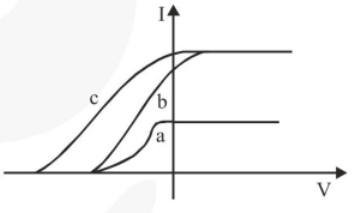
(A) none

(B) A only

(C) A and B only

(D) all the three metals.

Q7. The figure shows the variation of photocurrent with anode potential for a photosensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b and f_c be the frequencies for the curves a, b and c respectively :



(A) $f_a = f_b$ and $I_a \neq I_b$

(B) $f_a = f_c$ and $I_a = I_c$

(C) $f_a = f_b$ and $I_a = I_b$

(D) $f_b = f_c$ and $I_b = I_c$

Q8. The threshold frequency for a certain metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz is incident on the metal, the cutoff voltage for the photoelectric emission is.

(A) 8 V (B) 6 V

(C) 2 V (D) 4 V

Q9. Light coming from a discharge tube filled with hydrogen falls on the cathode of the photoelectric cell. The work function of the surface of cathode is 4eV. Which one of the following values of the anode voltage (in Volts) with respect to the cathode will likely to make the photo current zero?

(A) – 4 (B) – 6

(C) – 8 (D) – 10

Q10. Two particles have identical charges. If they are accelerated through identical potential differences, then the ratio of their deBroglie wavelengths would be :

(A) $\lambda_1 : \lambda_2 = 1 : 1$

(B) $\lambda_1 : \lambda_2 = m_2 : m_1$

(C) $\lambda_1 : \lambda_2 = \sqrt{m_2} : \sqrt{m_1}$

(D) $\lambda_1 : \lambda_2 = \sqrt{m_1} : \sqrt{m_2}$

Q11. The energy that should be added to an electron to reduce its De broglie wavelength from 10^{-10}m to 0.5×10^{-10} m will be :

(A) Four times the initial energy

(B) Equal to initial energy

(C) Twice the initial energy

(D) Thrice the initial energy

Q12. If the mass of neutron = 1.7×10^{-27} kg. then the De broglie wavelength of neutron of energy 3eV is :

(A) 1.6×10^{-10} m

(B) 1.6×10^{-11} m

(C) 1.4×10^{-10} m

(D) 1.4×10^{-11} m

Q13. A proton and an α–particle have kinetic energy in the ratio 16:1. The ratio of de-Broglie wavelength associated with them is:-

(A) 4 : 1 (B) 2 : 1

(C) 1 : 2 (D) 7 : 2

Q14. The mass of a particle is m. If its kinetic energy is increased to four times, then de-Broglie wavelength will :

(A) Remain unchanged

(B) Become half

(C) Become double

(D) Become $\frac{1}{\sqrt{2}}$ times.

Q15. The energy of a photon is equal to the kinetic energy of a proton. The energy of the photon is E. Let λ_1 be the de–Broglie wavelength of the proton and λ_2 be the wavelength of the photon. The ratio λ_1/λ_2 is proportional to :-

(A) E° (B) $E^{1/2}$

(C) E^{-1} (D) E^{-2}

Q16. A neutron having kinetic energy 13.6 eV collides with a hydrogen atom in ground state at rest. Assume that the mass of neutron and hydrogen atoms are same and the neutron does not leave its line of motion. Then which of the following is the possible KE of the neutron after the collision?

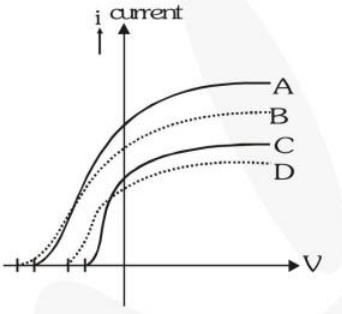
(A) zero (B) 3.4 eV

(C) 1.5 eV (D) 6.8 eV.

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The figure shows the results of an experiment involving photoelectric effect. The graphs A, B, C and D relate to a light beam having different wavelengths. Select the correct alternative.



(A) Beam B has highest frequency

(B) Beam C has longest wavelength

(C) Beam A has highest rate of photoelectric emission

(D) Photoelectrons emitted by B have highest momentum

Q2. In a photoelectric experiment, the collector plate is at 2.0V with respect to the emitter plate made of copper (ϕ= 4.5 eV). The emitter is illuminated by a source of monochromatic light of wavelength 200 nm.

(A) The minimum kinetic energy of the photoelectrons reaching the collector is 0.

(B) The maximum kinetic energy of the photoelectrons reaching the collector is 3.7 eV.

(C) If the polarity of the battery is reversed then answer to part A will be 0

(D) If the polarity of the battery is reversed then answer to part B will be 1.7 eV

Q3. Cut off potential for a metal in photoelectric effect for light of wavelength λ_1, λ_2 and λ_3 is found to be V_1, V_2 and V_3 volts if V_1, V_2 and V_3 are in Arithmetic Progression and λ_1, λ_2 and λ_3 will be :

(A) Arithmetic Progression

(B) Geometric progression

(C) Harmonic Progression

(D) None

Q4. In a photoelectric effect experiment, if f is the frequency of radiations incident on the metal surface and I is the intensity of the incident radiations, then choose the correct statement(s).

(A) If f is increased keeping I and work function constant then maximum kinetic energy of photoelectron increases.

(B) If distance between cathode and anode is increased stopping potential increases.

(C) If I is increased keeping f and work function constant then stopping potential remains same and saturation current increases.

(D) Work function is decreased keeping f and I constant then stopping potential increases.

Q5. The collector of the photocell (in photo-electric experiment) is made of tungsten while the emitter is of Platinum having work function of 10 eV. Monochromatic radiation of wavelength 124 Å & power 100 watt is incident on emitter which emits photo electrons with a quantum efficiency of 1%. The accelerating voltage across the photocell is of 10,000 volts (Use : hc = 12400eV Å)

(A) The power supplied by the accelerating voltage source is 100 watt

(B) The minimum wavelength of radiation coming from the tungsten target (collector) is 1.23 Å

(C) The power supplied by the accelerating voltage source is 10 watt

(D) The minimum wavelength of radiation coming from the tungsten target (collector) is 2.23 Å

Q6. Light of wavelength λ_1 & λ_2 are falling on two metal surface A & B. For wavelength λ_1 electron ejected from both the surfaces and for wavelength λ_2 electron ejected from only surface B. On the basis of these facts which of the following is/are false statement (s).

(A) more energy is required for ejection of electron from metal 'A'

(B) $\lambda_1 > \lambda_2$

(C) threshold wavelength for A is greater than B

(D) energy of electron ejected from metal A will be greater than electron ejected from metal B for a given wavelength λ

Q7. The accelerating potential (V) applied between a photocathode and the respective anode is such that the fastest photoelectron can fly only one fourth of the distance between the cathode and the anode. If the distance between photocathode and anode is reduced to $(1/4)^{th}$ of the original value while maintaining the accelerating potential constant, then -

(A) The fastest electron will reach the anode.

(B) The fastest electron will reach upto one fourth of the new distance between the cathode and the anode.

(C) Kinetic energy of the fastest photo-electron emitted from photocathode will not change due to change in plate separation (keeping V constant).

(D) Kinetic energy of the fastest photo-electron will increase due to decrease in plate separation (keeping V constant).

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No 8 to 10

A mercury arc lamp provides 0.1 watt of ultra-violet radiation at a wavelength of λ = 2537 Å only. The photo tube (cathode of photo electric device) consists of potassium and has an effective area of 4 cm^2. The cathode is located at a distance of 1m from the radiation source. The work function for potassium is ϕ_0 = 2.22 eV.

Q8. According to classical theory, the radiation from arc lamp spreads out uniformly in space as spherical wave. What time of exposure to the radiation should be required for a potassium atom (radius 2Å) in the cathode to accumulate sufficient energy to eject a photo-electron ?

(A) 352 second (B) 176 second

(C) 704 seconds (D) No time lag

Q9. To what saturation current does the flux of photons at the cathode corresponds if the photo conversion efficiency is 5%.

(A) 32.5 nA (B) 10.15 nA

(C) 65 nA (D) 3.25 nA

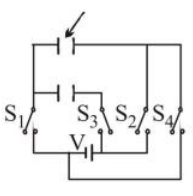
Q10. What is the cut off potential V_0 ?

(A) 26.9 V (B) 2.69 V

(C) 1.35 V (D) 5.33 V

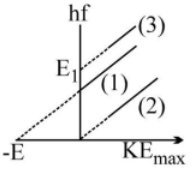
Paragraph for Question No 11 to 13

The circuit shown is placed in vacuum. Both the capacitors are identical and they have the same capacitance C. Light is incident on the left plate of the upper capacitor. When all the switches are open then the hf versus KE_{max} is shown by the straight line (1). In all the cases, we are measuring the KE_{max} when the electron reaches the opposite plate.



When only S_1 the switches and S_2 are closed, the graph becomes (2).

When only S_3 and S_4 are closed then the graph becomes (3).



Q11. What is the work function of the cathode?

(A) E (B) E_1

(C) $E + E_1$ (D) none of these

Q12. What is the value of eV ?

(A) E (B) E_1

(C) $E + E_1$ (D) none of these

Q13. What is the value of E_1?

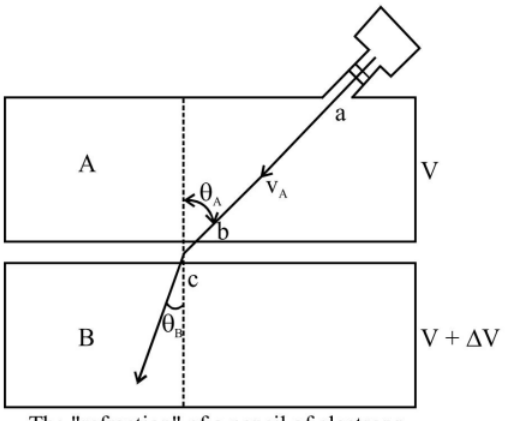
(A) 3E (B) 3E/2

(C) E/2 (D) none of these

Paragraph for Question No 14 to 16

Let a pencil of electrons from a suitable gun G enter through orifice in an enclosed metal box A, which has potential V relative to filament of the gun. Let these electrons emerge from A through orifice b and enter through c to another box B, which is maintained at a potential V + ΔV. The electric field between the two boxes changes the component of velocity of the electrons perpendicular to the adjacent surface, and the electrons enter B with a change in their direction of motion. Let v_A and v_B be the velocities of the electrons in A and B, respectively, and θ_A, θ_B the angles between these directions and the normal to the box faces at b and c. Since the electric field does not change the horizontal component of velocity, $v_A \sin\theta_A = v_B \sin\theta_B$,

$\frac{\sin\theta_A}{\sin\theta_B} = \frac{v_B}{v_A}$.



The "refraction" of a pencil of electrons

Now if we dealing with light waves undergoing refraction, or any other kind of wave, the relation would be :

$$\frac{\sin\theta_A}{\sin\theta_B} = \mu = \frac{u_A}{u_B}$$

where μ is the relative refractive index of the two media and u_A, u_B are the corresponding velocities of light wave. Comparison of the last two equations gives the result $\frac{u_A}{u_B} = \frac{v_B}{v_A}$. We may conclude that if matter waves follow the electron along its path, the wave speed u is inversely proportional to the speed v of the electron, or $u = \frac{b}{v}$.

Q14. If we define the frequency of matter waves as $f = \frac{u}{\lambda}$, the -

(A) Frequency of matter waves in medium A is more than that in medium B

(B) Frequency of matter waves in medium A is less than that in medium B

(C) Frequency of matter waves in medium A is same as that in medium B

(D) Cannot be predicted

Q15. Suppose $V_A = 20$ volt and $V_B = 15$ volt. Choose the **CORRECT** statement :-

(A) The speed of electrons as well as speed of matter waves inside box B is more

(B) The speed of electrons as well as speed of matter waves inside box B is less

(C) The speed of electrons inside the box B is more, but speed of matter waves in box B is less

(D) The speed of electrons inside the box B is less, but speed of matter waves in box B is more

Q16. The refractive index for matter waves can be defined as $\frac{c}{u}$ where c is some constant. So refractive index for matter waves

(A) is inversely proportional to λ

(B) is independent of λ

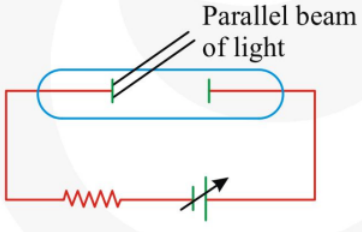
(C) is directly proportional of λ

(D) is proportional to $\sqrt{\lambda}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. In the shown experimental setup to study photoelectric effect, two conducting electrodes are enclosed in an evacuated glass-tube as shown. A parallel beam of monochromatic light, falls on photosensitive electrodes. The emf of battery shown is high enough such that all photoelectrons ejected from left electrode will reach the right electrode. Under initial conditions photoelectrons are emitted. As changes are made in each situation of column I; Match the statements in column I with results in column II.



Column I	**Column II**
(A) If frequency of incident light is increased keeping its intensity constant	(P) magnitude of stopping potential will increase
(B) If frequency of incident light is increased and its intensity is decreased	(Q) current through circuit may stop
(C) If work function of photo sensitive Electrode is increased	(R) maximum kinetic energy of ejected photoelectrons will increase
(D) If intensity of incident light is increased Keeping its frequency constant	(S) saturation current will increase

NUMERICAL TYPE

Q18. Threshold frequency of a surface is ν_0. It is illuminated by $3\nu_0$ frequency, then maximum speed of photo electrons is V m/sec. If incident frequency is $9\nu_0$, maximum speed is NV. Find N ?

Q19. When incident wavelength is λ, stopping potential is $3V_0$. If incident wavelength is 2λ then stopping potential is V_0. Threshold wavelength is $N\lambda$. Find N

Q20. In a photoelectric setup, the radiations from the Balmer series of hydrogen atom are incident on a metal surface of work function 2eV. The wavelength of incident radiations lies between 450 nm to 700 nm. Find the maximum kinetic energy of photoelectron emitted.
(Given hc/e = 1242 eV-nm).

SUBJECTIVE TYPE

Q21. A parallel beam of monochromatic light of wave length 500 nm is incident normally on a perfectly absorbing surface. The power through any cross-section of the beam is 10 W.
Find :
(a) Number of photon absorbed by the surface per second.
(b) The force exerted by light beam on the surface.

Q22. Calculate number of photons passing through a ring of unit area in unit time if light of intensity 100 Wm^{-2} and of wavelength 400 nm is falling normally on the ring.

Q23. A monochromatic light incident on metal 'A' having threshold frequency ν_0. It emits photo electrons of maximum kinetic energy K. Now incident frequency is made three times and fall on a metal 'B' having threshold frequency $2\nu_0$. What will be the maximum kinetic energy of photo electrons emitted by metal 'B' ?

Q24. A particle moving with velocity that is three times that of velocity of electron. If ratio of de-Broglie wave length of particle to that of the electron is 1.8×10^{-4}.
Find mass of particle ($m_e = 9.1 \times 10^{-31}$ kg)

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	C	B	B	C	A	B	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**			
A	B	C	C	C	D	B			

Ex-1A

1. A,C **2.** B,C,D **3.** D **4.** A,B,C **5.** A,C

6. D **7.** A,B,C **8.** (A) R,S (B) P,Q (C) P,Q (D) R,S

9. 1.7 eV **10.** 6 **11.** No **12.** 1.5×10^{34} **13.** 1:1

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	B	D	C	B	C	A	C	D	C
11	**12**	**13**	**14**	**15**	**16**				
D	B	C	B	B	A				

Ex-2A

1. A,B,C,D **2.** B **3.** C **4.** A,C,D **5.** A,B

6. B,C,D **7.** B,C **8.** A **9.** A **10.** B

11. A **12.** A **13.** B **14.** C **15.** D

16. A **17.** (A)-P,R (B)-P,R (C)-Q (D)-S **18.** 2

19. 4 **20.** 0.55 eV 21. (a) 2.5×10^{19} (b) 3.33×10^{-7} N

22. 2.02×10^{20} **23.** $3K + h\nu_0$ 24. 1.68×10^{-27} kg